INCH-POUND
MIL-PRF-10304/17F
w/AMENDMENT 1
18 January 2008
SUPERSEDING
MIL-PRF-10304/17F
26 April 1999

PERFORMANCE SPECIFICATION SHEET

METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: AMMETER, DC (FLUSH MOUNTING, ROUND FLANGE, 4.5 INCH), STYLE 46

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-10304.

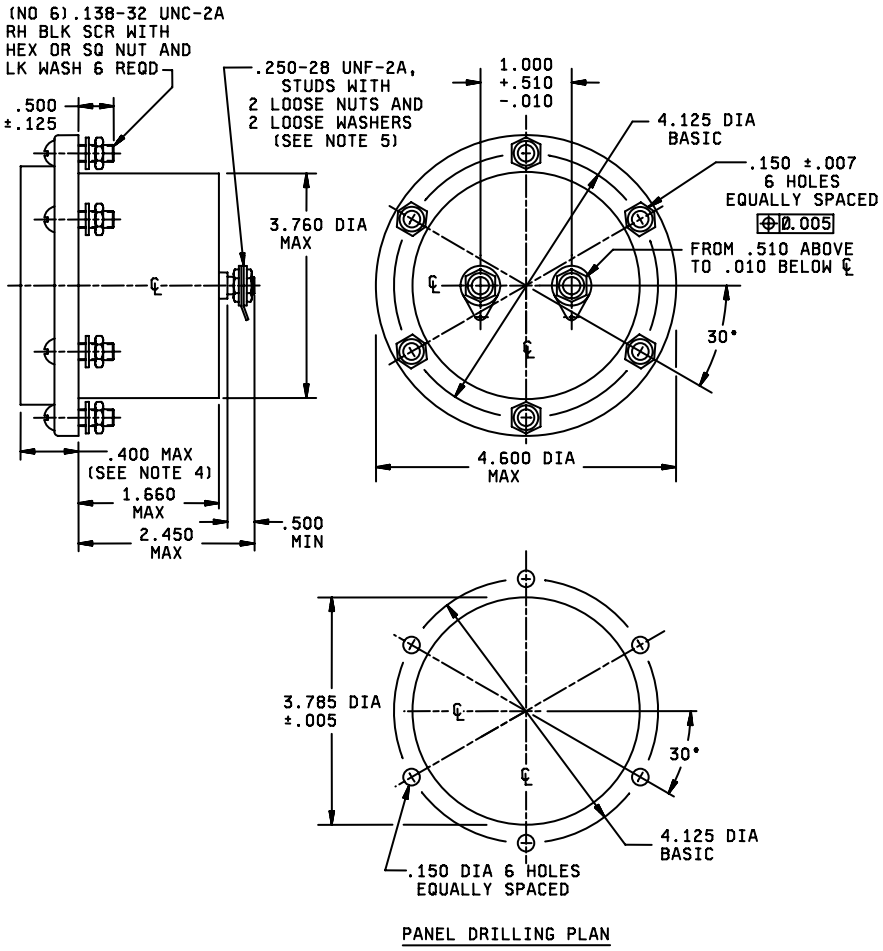


FIGURE 1. Meter, panel type.

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Inches	mm	Inches	mm	Inches	mm
.005	.13	.250	6.35	2.450	62.23
.007	.18	.400	10.16	3.760	95.50
.010	.25	.500	12.70	3.785	96.14
.125	3.18	.510	12.95	4.125	104.78
.138	3.51	1.000	25.40	4.600	116.84
.150	3.81	1.660	42.16		

NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 inch for two place (.XX), and ± .005 inch for three place (.XXX) dimensions.
2. Tolerance is ± 0.5 (½) degree on angles.
3. Metric equivalents are given for general information only.
4. Maximum projection from front panel including mounting gaskets (if used).
5. Other forms of terminals, as specified in MIL-PRF-10304, are permitted. The specified minimum length applies to stud-type terminals only. Terminals shall be equally spaced, from each side of vertical centerline.

FIGURE 1. Meter, panel type – Continued.

REQUIREMENTS:
(Readings expressed as a percent of full-scale value.)

Dimensions and configuration: See figure 1.

Full-scale ranges: See table I.

Scale:
Length: 3.25 inches minimum.
Pointer deflection: 90 degrees minimum.

Position influence: ± 2 percent change of 60 degrees rotation from normal operating position.

Accuracy: ± 2 percent.

Overshoot: See table I.

Response time: See table I.

Power consumption (loss) (at end scale deflection): See table I.

High temperature cycling:
± 4 percent.
± 2 percent permanent change.

Temperature influence: ± 1 percent change.
(Type MR46W001DCMAR meters shall be tested as millivoltmeters in this test.)

Exposure to extreme temperatures: ± 3 percent permanent change.
(Type MR46W001DCMAR meters shall be tested as millivoltmeters in this test.)

Overload capacity:
Momentary overload: ± 2 percent permanent change.

Sustained overload:
± 1 percent temporary zero shift.
± 1 percent permanent zero shift.
± 2 percent permanent change.

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Dielectric withstanding voltage: 3,000 volts rms.

Vibration, random drop, and shock: Repeatability is not applicable for those microammeters below 100-microampere ranges.

TABLE I. Type designation and characteristics.

Type designation	Full-scale ranges		Overshoot (max)	Resistance	Power consumption (loss)	Response time
	Zero	Range	%	ohms	millivolts	seconds max
MR46W001DCMAR	Left	1 mA 100 mV	40	100 nom	100 ± 2	2
MR46W050DCMAR	Left	50 μA	50	2,500 ± 2%	125 nom	3

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: This document references MIL-PRF-10304.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 99
DLA – CC

Preparing activity:
DLA – CC

(Project 6625-2006-014)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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